



Politechnika Krakowska
im. Tadeusza Kościuszki

IPv4 ADDRESSING

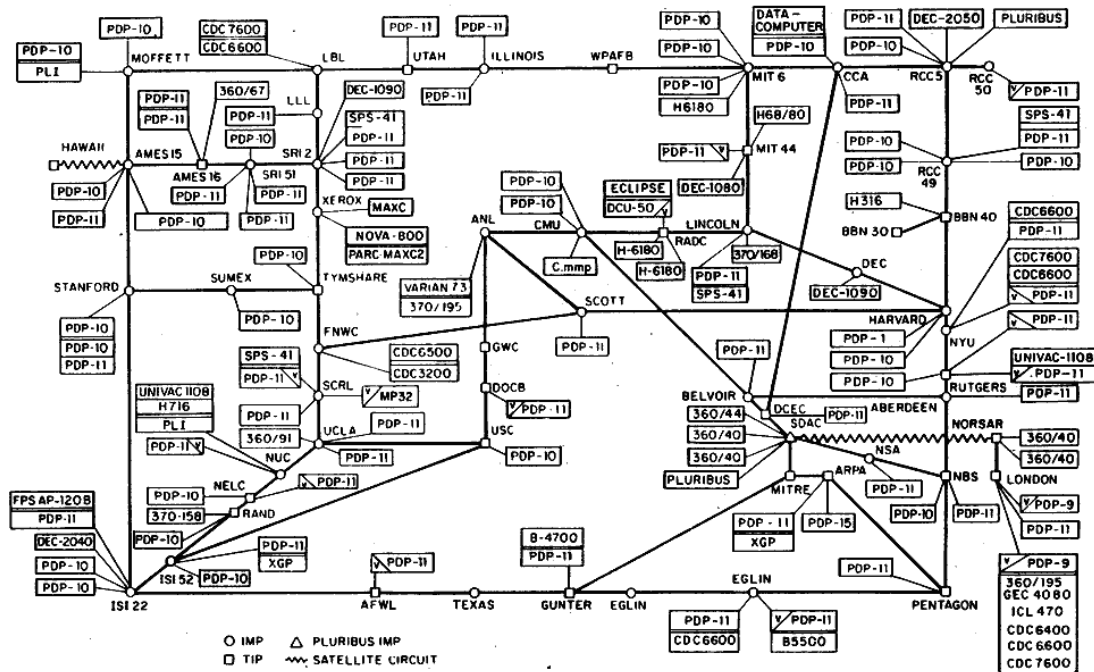
dr inż. Paweł Król
Katedra Automatyki i Informatyki (E-1)
Politechnika Krakowska

Protocol wars

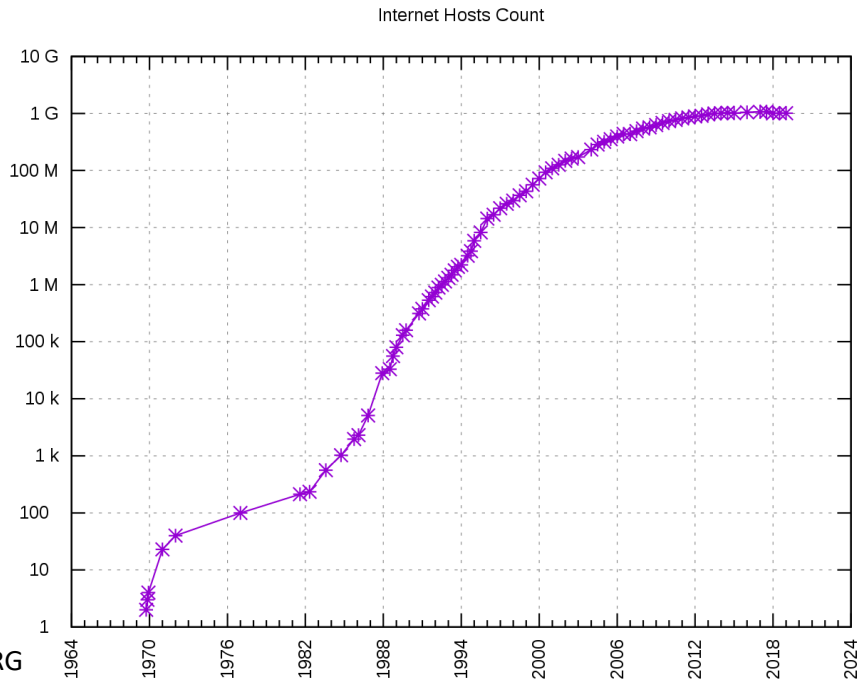
1977 >>

(PLEASE NOTE THAT WHILE THIS MAP SHOWS THE HOST POPULATION OF THE NETWORK ACCORDING TO THE BEST INFORMATION OBTAINABLE, NO CLAIM CAN BE MADE FOR ITS ACCURACY)

NAMES SHOWN ARE IMP NAMES, NOT (NECESSARILY) HOST NAMES



Development of internet network



Źródło: WIKIPEDIA.ORG

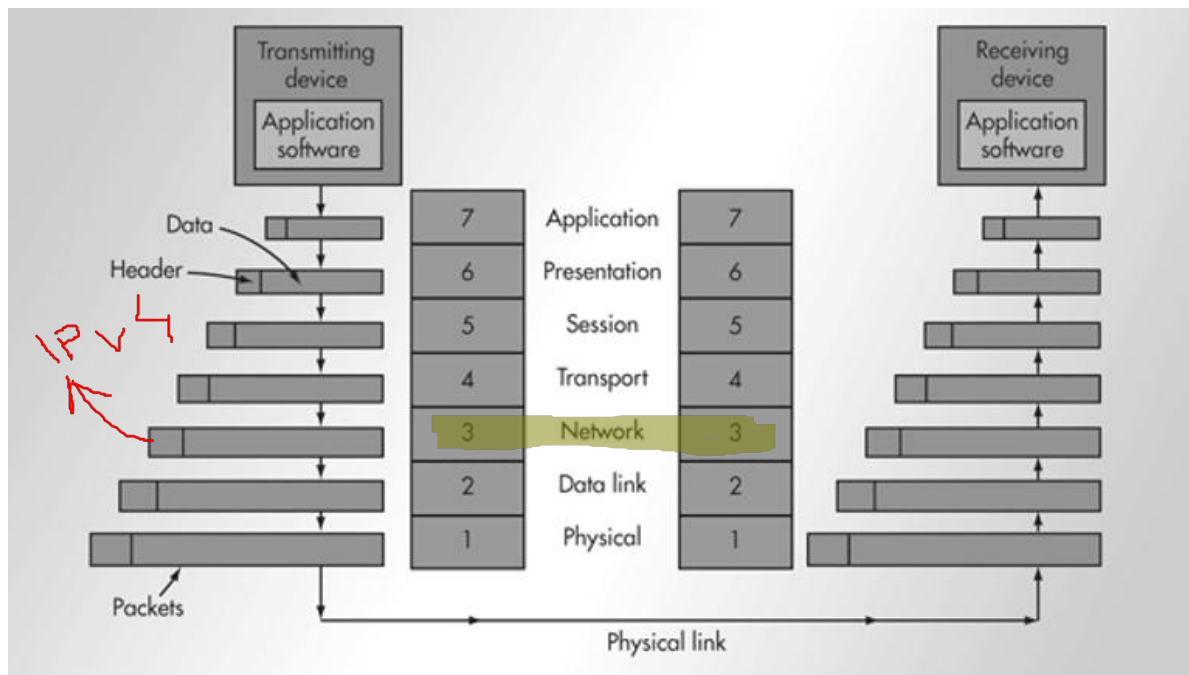
Protocol wars

- **1969** the ARPANET network was created - beginnings of networks
- **Before 1981** in IP addresses only the first 8 bits represented the network,
- **Since 1981** A 32-bit IP address allows the use of three classes (RFC 791),
- **Before 1983** – NCP (Network Control Protocol)
- **1 | 1983** – ARPANET changed its core networking protocols from NCP to the more flexible and powerful TCP/IP („a flag day”)
- **1993** classless inter-domain routing created CIDR (RFC 1519),

https://en.wikipedia.org/wiki/Protocol_Wars

[https://en.wikipedia.org/wiki/Flag_day_\(computing\)](https://en.wikipedia.org/wiki/Flag_day_(computing))

IPv4 addresses



Network layer

Network layer:

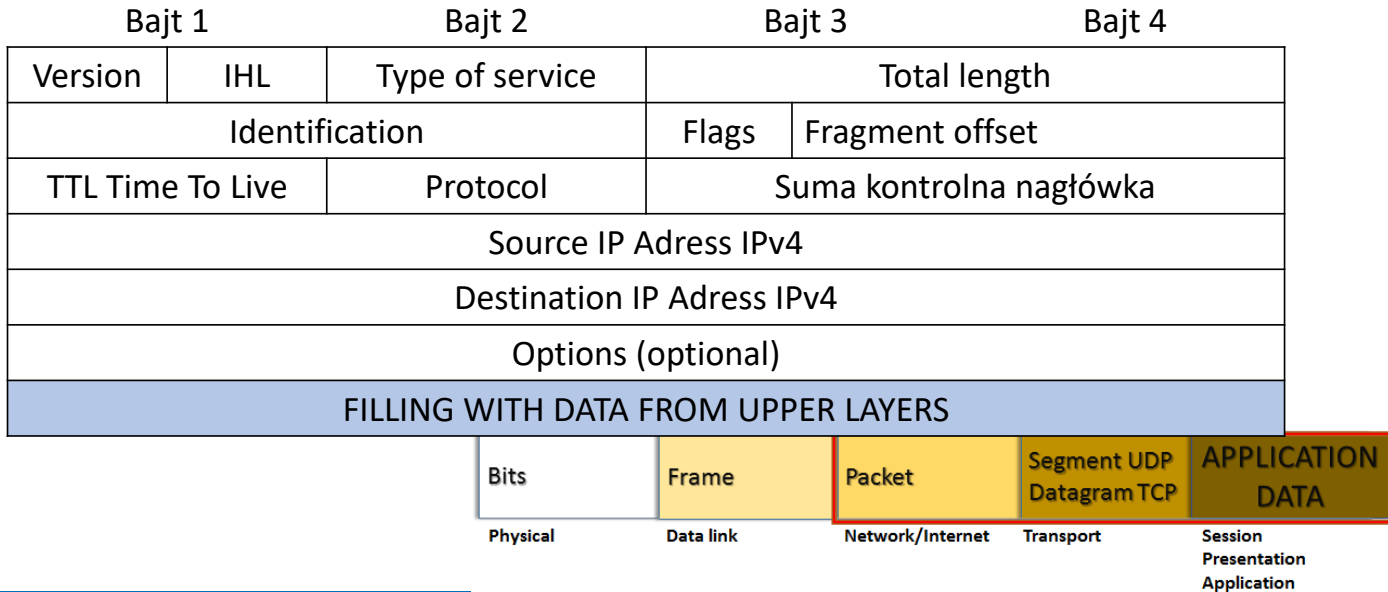
- Provides a connection between the sender and receiver hosts,
- Each sending and receiving device has a unique address (e.g. IPv4),

The network layer performs the key task of routing in communication between networks – finding the best route for a packet in the network.

Network protocols

	MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation hardware	7 Application				HTTP, DNS, SMTP, POP, SSL, SQL, DHCP	Application
	6 Presentation					
	5 Session					
	4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
	3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
	2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
	1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

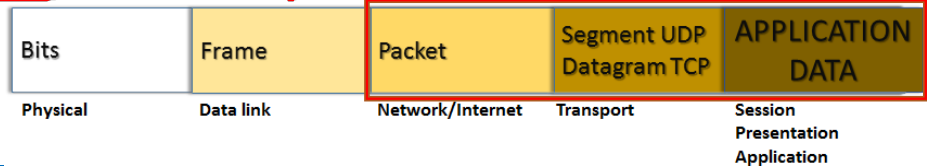
IPv4 header



Adres IPv4, Network mask

IP address 192.168.1.45			
192	168	1	45
11000000	10101000	00000001	00101101

Network mask 255.255.255.0 or ____/24			
255	255	255	0
11111111	11111111	11111111	00000000



Class addressing

class of
IP address
defines
network
mask

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	<u>0.0.0.0 - 127.255.255.255</u>	<u>255.0.0.0</u>	127	16.777.214
B	10 _ _	<u>128.0.0.0 - 191.255.255.255</u>	255.255.0.0	16.382	65.534
C	110 _	<u>192.0.0.0 - 223.255.255.255</u>	<u>255.255.255.0</u>	2.097.150	254
D	1110	<u>224.0.0.0 - 239.255.255.255</u>	for multicast transmission		
E	1111	<u>240.0.0.0 - 255.255.255.255</u>	Reserved for IETF		

Adresy localhost (loopback)

IP: 127.0.0.1

<https://en.wikipedia.org/wiki/Localhost>

```
H:\>ping localhost
```

```
Pinging PK [::1] with 32 bytes of data:
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Ping statistics for ::1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
H:\>ping 127.0.0.1
```

```
Pinging 127.0.0.1 with 32 bytes of data:
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 127.0.0.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
H:\>
```

Address 0.0.0.0

0.0.0.0

- Undefined address IP

0.0.0.0/0

- IP block containing all IP addresses
- This is how a static route is marked

<https://en.wikipedia.org/wiki/0.0.0.0>

IPv4 link-local

automatic private IP addressing (APIPA)

- 169.254.0.0/16 -> 169.254.0.0 – 169.254.255.255

Local addresses



class	Range of IP addresses	Network mask	CIDR notation
A	10.0.0.0 – 10.255.255.255	255.0.0.0	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	255.240.0.0	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	255.255.0.0	192.168.0.0/16

<https://whatismyipaddress.com/private-ip>

IPv4 multicast

Selected IP addresses of the **class D** (range 224.0.0.0 - 239.255.255.255)

- 224.0.0.0/24 only used to connect to a local multicast router
- 232.0.0.0/8 is reserved for multicast from the selected source (SSM).
- 233.0.0.0/8 is disconnected from addresses GLOP

Reservation of all Class D IP addresses:

<https://www.iana.org/assignments/multicast-addresses/multicast-addresses.xhtml>

Adres IPv4 – types of broadcast

Directed broadcast IPv4 address

- Has a specific network part, and in the host part it has all ones,
- If the network address is known - 172.16.255.255/16
- Used in ARP and DHCP.

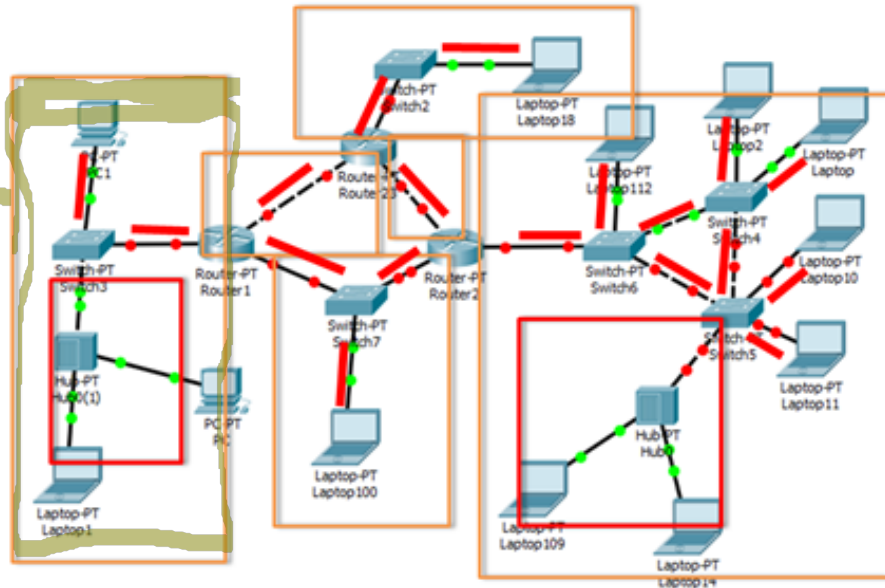
Limited broadcast IPv4 address

- An address where all 32 bits are set to one – 255.255.255.255,
- In case the network address is unknown,

Broadcast domains vs collision domains

Broadcast domains
(domeny rozgłoszeniowe)

Collision domains
(domeny kolizji)



IPv4 ADDRESSING – tasks

Converting a mask to slash notation

What is the slash form of a decimal mask?

• $\frac{255}{8}.\frac{255}{8}.0.0 \rightarrow /16$

• $\frac{255}{8}.\frac{255}{8}.\frac{192}{2}.0 \rightarrow /18$

• $\frac{255}{8}.\frac{255}{8}.\frac{255}{8}.\frac{240}{4} \rightarrow /28$

• $\frac{255}{8}.\frac{255}{8}.\frac{255}{8}.\frac{252}{4} \rightarrow /30$

• $\frac{255}{8}.\frac{255}{8}.\frac{255}{8}.\frac{254}{2} \rightarrow /31$

Przeliczanie maski do postaci dziesiętnej

What is the decimal form of a slash mask?

- $______ / 24 \rightarrow 255.255.255.0$

- $______ / 22 \rightarrow 255.255.252.0$

- $______ / 23 \rightarrow 255.255.254.0$

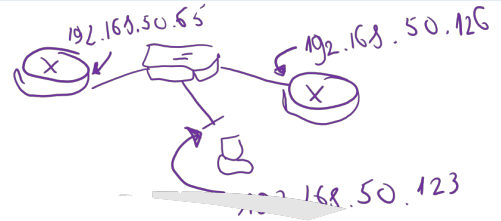
- $______ / 25 \rightarrow 255.255.255.128$


Identification I

IP address: **192.168.50.123**

Net mask: **255.255.255.192** or /

A	B	C
0	0	0
1	1	0
1	0	1
1	1	0



IP address:

1100 0000 . 1010 1000 . 0011 0010 . 01 | **11 1011**

Net mask :

1111 1111 . 1111 1111 . 1111 1111 . 11 | **00 0000**

Network addr.:

1100 0000 . 1010 1000 . 0011 0010 . 01 | **00 0000**

First host addr.:

1100 0000 . 1010 1000 . 0011 0010 . 01 | **00 0001** +1

..

Last host addr:

1100 0000 . 1010 1000 . 0011 0010 . 01 | **11 1110**

broadcast addr:

1100 0000 . 1010 1000 . 0011 0010 . 01 | **11 1111** -1

Provide:

- Network address: **192.168.50.64**
- First host address: **192.168.50.65**
- The last host address: **192.168.50.126**
- Broadcast address: **192.168.50.127**
- The number of hosts: $2^{(\text{zeros_in_the_net_mask})} - 2 = 2^6 - 2 = 64 - 2 = 62$

CIDR, VLSM

Class addressing

*this standard
used before CIDR*

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	_ 0.0.0.0 - <u>127</u> .255.255.255	<u>255.0.0.0</u>	127	16.777.214
B	<u>10</u> _ _	<u>128</u> .0.0.0 - <u>191</u> .255.255.255	<u>255.255.0.0</u>	16.382	65.534
C	<u>1110</u> _	<u>192</u> .0.0.0 - <u>223</u> .255.255.255	<u>255.255.255.0</u>	2.097.150	254
D	1 1 1 0	<u>224</u> .0.0.0 - <u>239</u> .255.255.255	<u>for multicast transmission</u>		
E	1 1 1 1	<u>240</u> .0.0.0 - <u>255</u> .255.255.255	<u>Reserved for IETF</u>		

CIDR (Classless Inter-Domain Routing)

CIDR (Classless Inter-Domain Routing)

- Classless IP routing method between domains
- introduces a variable mask length (VLSM), which enables network aggregation and division of the network into subnets,
- Introduced in 1993 by the IETF (Internet Engineering Task Force) to replace the earlier, classful addressing method,

Number of subnetworks for network in class B

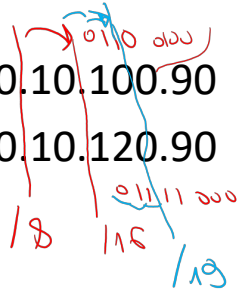
Network address: 172.16.0.0 | Network mask: 255.255.0.0 $\Leftrightarrow$ __.__.__.__./16

Network address	Network mask		Number of subnets	Number of host addresses
172.16.0.0	255.255.192.0	/18 .11 000000.00000000	2	16.382
172.16.0.0	255.255.224.0	/19 .111 00000.00000000	6	8.190
172.16.0.0	255.255.240.0	/20 .1111 0000.00000000	14	4.094
172.16.0.0	255.255.248.0	/21 .11111 000.00000000	30	2.046
172.16.0.0	255.255.252.0	/22 .111111 00.00000000	62	1.022
172.16.0.0	255.255.254.0	/23 .1111111 0.00000000	126	510
172.16.0.0	255.255.255.0	/24 .11111111. 00000000	254	254
172.16.0.0	255.255.255.128	/25 .11111111.1 00000000	510	126
172.16.0.0	255.255.255.192	/26 .11111111.11 00000000	1.022	62
172.16.0.0	255.255.255.224	/27 .11111111.111 00000000	2.046	30
172.16.0.0	255.255.255.240	/28 .11111111.1111 00000000	4.094	14
172.16.0.0	255.255.255.248	/29 .11111111.11111 00000000	8.190	6
172.16.0.0	255.255.255.252	/30 .11111111.111111 00000000	16.382	2

VLSM - tasks

Suggest a suitable network for hosts

- 10.10.100.90
- 10.10.120.90



10.10.100.90
10.10.120.90
10.10.224.0



decimal	Binary	
10.10.100.90	? 10.10.01101000,90	First host address
10.10.120.90	? 10.10.01111000,90	Second host address
?	? 255.255.224.0	Net mask
?	? 10.10.96.0	Network address

